

JOÃO JOSÉ

Senior C++ Software Engineer | Python | Systems Integration

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PROFESSIONAL SUMMARY

Senior C++ software engineer with 10+ years building and modernising business-critical systems in air traffic management, electronic trading and financial risk. Hands-on expertise in C++, Python, distributed integrations, messaging, APIs and databases, with international Agile delivery experience. Based in Portugal and available for fully remote international roles and contracts.

TECHNICAL SKILLS

Languages: C++, Python, C, Go, Ada, Bash

Integration & Messaging: REST APIs, Apache Kafka, Protobuf, ZeroMQ, PostgREST, RPC

Data & Identity: PostgreSQL, Ingres, SQL, LDAP, JWT, OAuth 2.0, Microsoft Entra ID

Testing & Diagnostics: GTest/GMock, Catch2, pytest, Valgrind, ASan/TSan/UBSan

Development Tools: Git, CMake, GCC, Clang, GDB, clang-tidy, Cppcheck

Platforms & CI/CD: Linux, Docker, Podman, Kubernetes, OpenShift, Jenkins, SonarQube, Ansible

PROFESSIONAL EXPERIENCE

Independent Software Engineer | JTJ Software

Nov 2022 - Present | Netherlands & Portugal

Portuguese Real-Estate Software

Internal product | Jan 2026 - Present | Portugal

- Designing and developing a software product for the Portuguese real-estate market, from product discovery and architecture through hands-on implementation.

SIM Builder

End client: LVNL (Dutch Air Traffic Control) | Jun 2024 - Feb 2026 | Netherlands

- Combined Scrum Master and hands-on developer responsibilities, coordinating delivery with the Product Owner and resolving technical and integration challenges.
- Contributed across the full stack of a web-based ATM scenario-building platform built on an adapted BlueSky simulator, including the browser UI, APIs, scenario generation, messaging, data persistence and containerised deployment.

Technologies: Python, Django, FastAPI, React, TypeScript, OpenLayers, PostgreSQL, ZeroMQ, Microsoft Entra ID, OAuth 2.0/PKCE, pytest, pytest-django, Docker, Nginx

iCAS SIM

End client: LVNL (Dutch Air Traffic Control) | Nov 2022 - Dec 2025 | Netherlands

- Developed and maintained iCAS SIM, the simulation and training environment supporting the Netherlands' next-generation air traffic management system.
- Integrated the simulator with multiple peripheral systems, diagnosing cross-system issues and supporting reliable end-to-end operation.

Technologies: C++, Ada, Python, Bash, PostgreSQL, Ingres, Docker, STL, Boost, CMake, GCC, GDB, GTK

C++ Software Developer | Accent Pointe B.V.

May 2021 - Nov 2022 | Netherlands

- Developed high-performance C++ trading software integrated with multiple brokers and market-data feeds, supporting real-time trader workflows.
- Modernised substantial areas of a legacy MFC and Microsoft RPC codebase, improving maintainability and enabling continued product development.

Technologies: C++, C, C#, Java, SQL, Bash, Python, MFC, RPC

C++ Software Developer | ABN AMRO Clearing Bank

Nov 2016 - May 2021 | Netherlands

- Delivered C++ software across multiple risk-management projects supporting business-critical clearing and financial workflows.
- Built and integrated service interfaces, messaging flows and authentication components supporting reliable communication between business-critical systems.

Technologies: C++, C, Go, C#, Java, Python, Bash, SQL, Ingres, REST APIs, PostgREST, ZeroMQ, Apache Kafka, Protobuf, LDAP, JWT, OpenShift, Podman

ADDITIONAL EXPERIENCE

Software Engineer | Frinch

Apr 2016 - Oct 2016 | Faro, Portugal / Remote

- Developed Product Information Management software in a distributed team collaborating with the company's headquarters in Munich.

Technologies: PHP, JavaScript

Computer Vision Researcher | Vision Laboratory, University of Algarve

Sep 2008 - Oct 2013 | Faro, Portugal

- Developed computer-vision algorithms and prototypes for assistive navigation, including real-time path, obstacle and lateral-door detection.
- Collaborated with the Institute for Systems and Robotics (ISR) and the University of Trás-os-Montes and Alto Douro (UTAD) on the FCT-funded SmartVision and Blavigator projects.
- Co-authored peer-reviewed publications on computer-vision-based navigation aids.

Cisco Instructor | Cisco Academy, FCT-UAIg

Nov 2008 - Jun 2009 | Faro, Portugal

- Module 1 - Network Fundamentals: network models and protocols, Ethernet, IP addressing and subnetting.
- Module 2 - Routing Protocols and Concepts: routing fundamentals, VLSM, RIP, OSPF and EIGRP.
- Module 3 - LAN Switching and Wireless: switch configuration, VLANs, trunking, spanning tree and WLANs.
- Module 4 - Accessing the WAN: WAN technologies, ACLs, PPP/PPPoE, Frame Relay and VPN concepts.

Invited Expert | Portuguese Immigration and Borders Service (SEF)

May 2007 - Aug 2007 | Faro, Portugal

- Participated in the quality assessment of the electronic border-control system installed at Faro Airport.

EDUCATION

Doctoral Research, Computer Science Engineering | University of Algarve

Sep 2011 - Oct 2013 | Portugal

- Conducted research into robust visual recognition of objects and environments for blind and visually impaired users.

MSc, Computer Science Engineering | University of Algarve

Sep 2008 - Jun 2011 | Portugal

- Thesis: Real-time path and obstacle detection for blind persons - grade 18/20.

BSc, Computer Science Engineering | University of Algarve

Sep 2003 - Jul 2008 | Portugal

CERTIFICATIONS & TRAINING

- Professional Scrum Master I (PSM I), Scrum.org - Mar 2018
- Certificate of Pedagogical Competences (CCP), IEFP - Mar 2015
- Cisco Certified Network Associate (CCNA), Cisco Academy FCT-UAIg - May 2008

LANGUAGES

Portuguese (Native) • English (Fluent) • Dutch (Intermediate) • Spanish (Good)